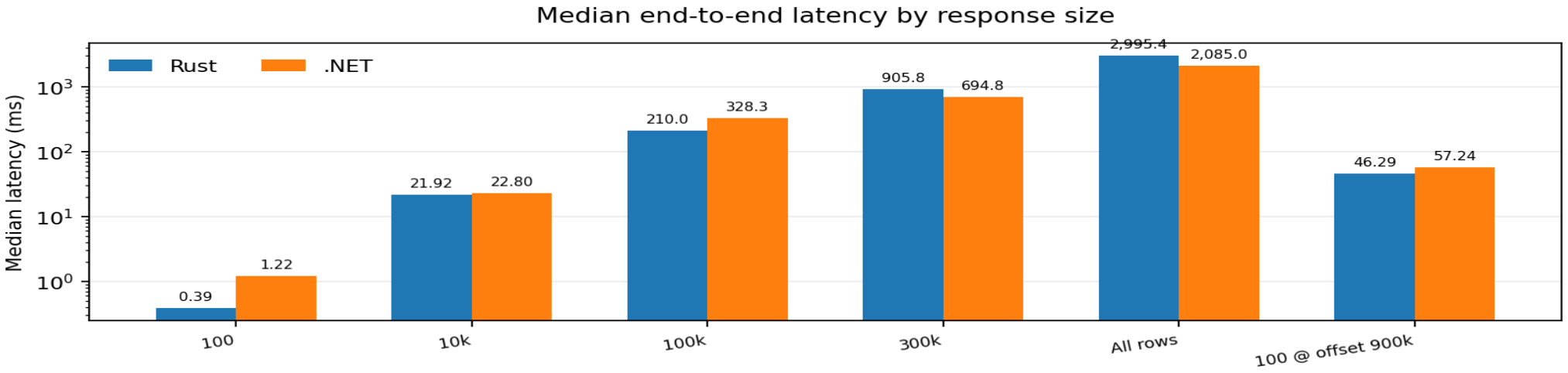
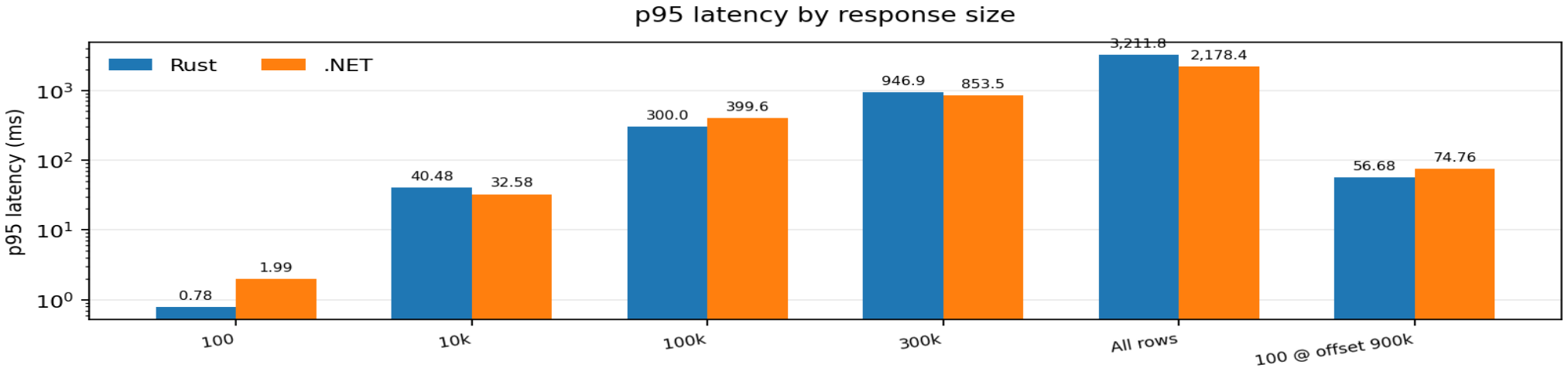
ORBIS BENCHMARK — TEST RUN 1

Rust (Axum) vs ASP.NET Core • SQLite • 1,000,003 products • values averaged across two rounds

1. Sequential request latency



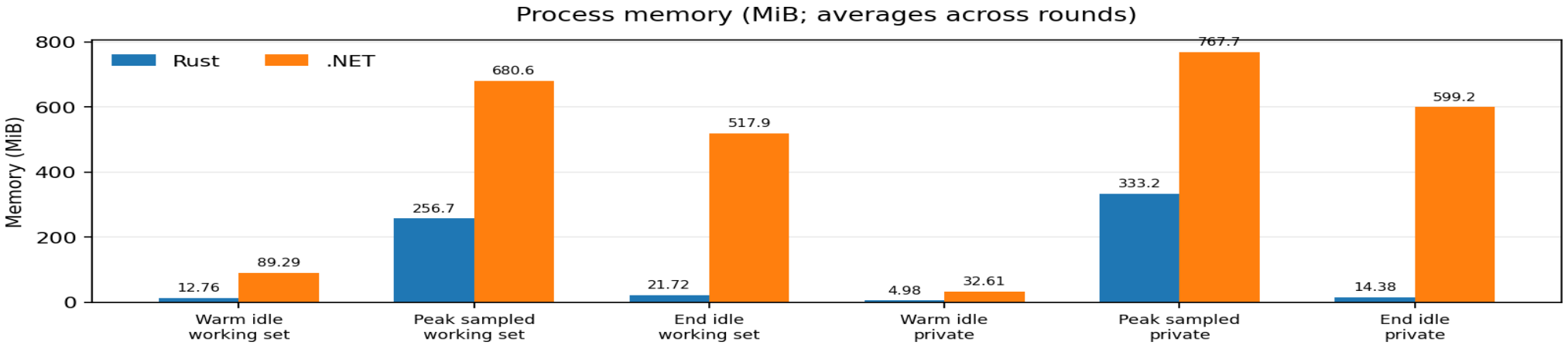


Timings include response transfer and client-side SHA-256 hashing, as described in the benchmark method.

ORBIS BENCHMARK — TEST RUN 1

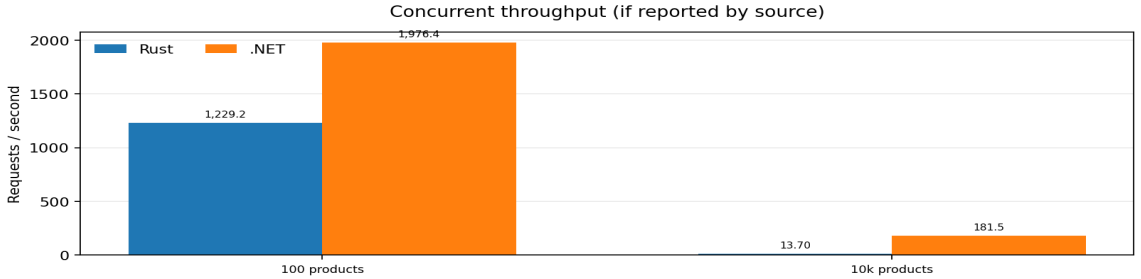
Resource use and concurrent request behaviour

2. Memory usage



Working set is resident process memory; private memory is process-private committed memory. Peak is sampled, not a guaranteed absolute maximum.

3. Concurrent requests (8 concurrent requests)



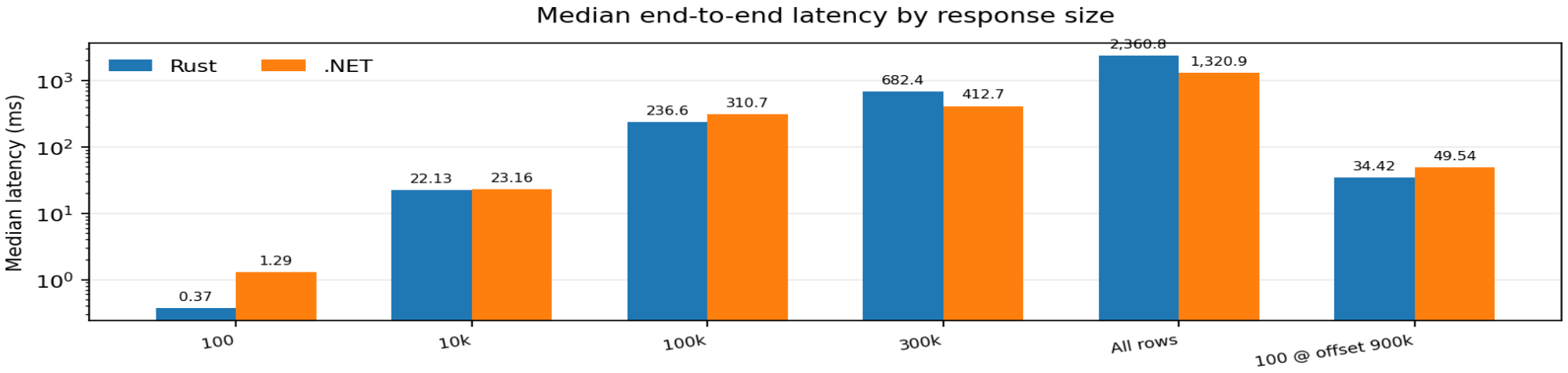
Stack	100 products median / p95 (ms)	10k products median / p95 (ms)
Rust	3.40 / 22.60	587.92 / 713.75
.NET	2.83 / 10.61	36.70 / 93.85

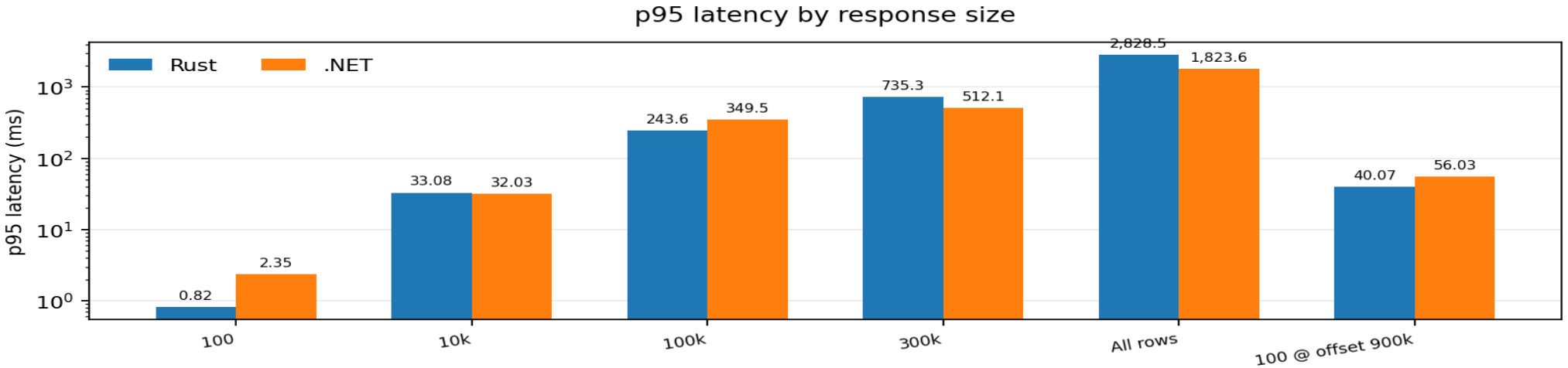
Note: concurrency throughput is taken from the source JSON when available; if absent, it is estimated from recorded sample count and elapsed duration. Large-payload concurrent results should be treated cautiously and validated.

ORBIS BENCHMARK — TEST RUN 2

Rust (Axum) vs ASP.NET Core • SQLite • 1,000,003 products • values averaged across two rounds

1. Sequential request latency



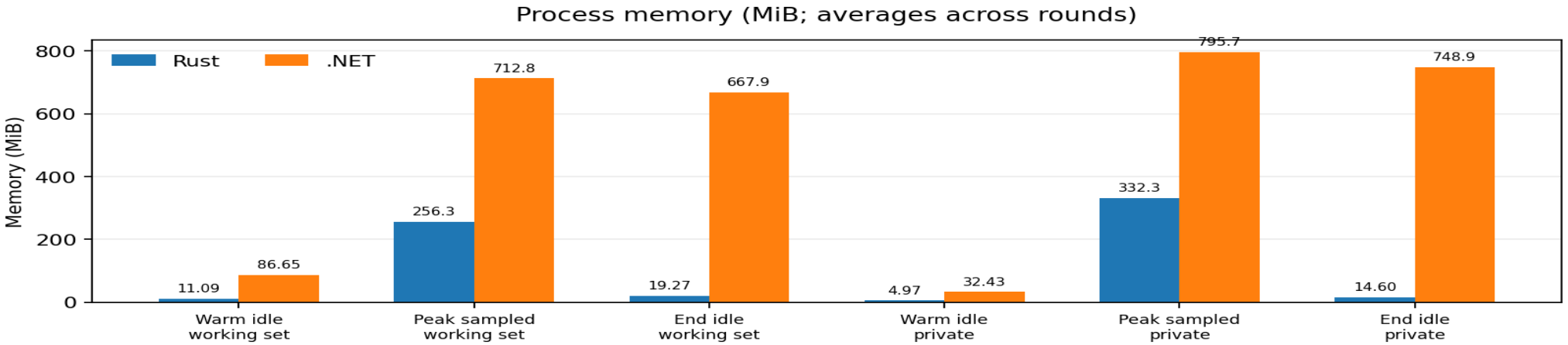


Timings include response transfer and client-side SHA-256 hashing, as described in the benchmark method.

ORBIS BENCHMARK — TEST RUN 2

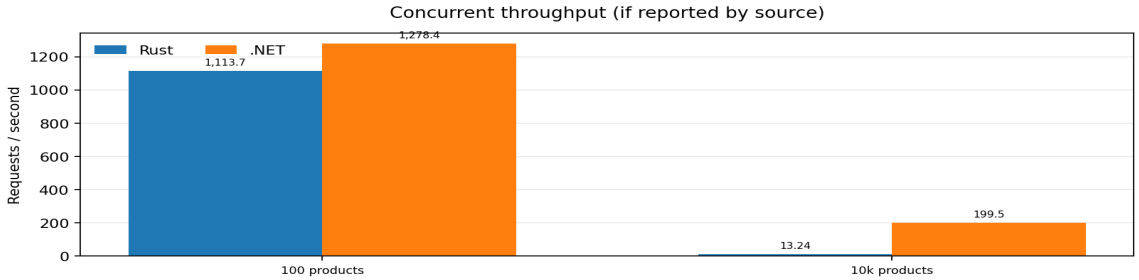
Resource use and concurrent request behaviour

2. Memory usage



Working set is resident process memory; private memory is process-private committed memory. Peak is sampled, not a guaranteed absolute maximum.

3. Concurrent requests (8 concurrent requests)



Stack	100 products median / p95 (ms)	10k products median / p95 (ms)
Rust	3.68 / 24.70	613.77 / 766.56
.NET	2.91 / 19.13	30.78 / 81.10

Note: concurrency throughput is taken from the source JSON when available; if absent, it is estimated from recorded sample count and elapsed duration. Large-payload concurrent results should be treated cautiously and validated.